



IDBI INNOVATE 2026

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Team Details

- a. **Team name:** ASM FinTech Developer
- b. **Team leader name:** Abhinav Mukwane
- c. **Problem Statement:** Financial Health Score

Brief about the idea

India has 6.4 crore MSMEs contributing ~30% of GDP, yet 80%+ have no formal credit. New-to-Credit (NTC) and New-to-Bank (NTB) enterprises are rejected for missing paperwork — not for being bad borrowers.

FinRiskLensAI — AI-powered MSME Financial Health Score platform (Top-5 Finalist build)

- One consent, five sources: Udyam, GST (7 return types), ITR, bank statements via Account Aggregator, and MCA company & director records
- One score, six pillars: Financial Health Score 0–1000 from a rules engine + ML.NET LightGBM — every score comes with plain-language reasons
- Financial Health Card: score gauge, 6-pillar radar, strengths & risks, lending eligibility (working capital + term loan) and the right product (CGTMSE / MUDRA / standard)
- **NEW — Score History:** every analysis is saved, so the bank sees the trend, not just today's number
- **NEW — Bank Portal (Customer 360):** one screen per MSME for the credit officer — score, GST, bank, ITR, MCA and directors
- Integrity check: GST, ITR and bank credits are cross-checked; mismatches are flagged for officer review, not silently scored
- Built for Bharat: 7 Indian languages (Bhashini), AI assistant, printable & emailable Financial Report; score in seconds; API-ready for LOS / ULI / OCEN / ONDC

Opportunities

How is it different from existing solutions?

- Bureau scores (CIBIL / Experian) need credit history — we score NTC / NTB MSMEs with none, from their own digital footprint
- Five data sources in one consent flow — Udyam, GST, ITR, bank (AA) and MCA — instead of a single-source view
- Score + score history + bank portal: a working system a credit officer can use today, not just a number
- Cross-source check (GST vs ITR vs bank credits) as a built-in integrity signal — no existing tool does this

How does it solve the problem?

- Ends the 'no documents, no credit' cycle — the MSME's existing filings and statements are the documents
- 2–3 weeks of manual appraisal becomes an automated assessment in seconds
- Every score carries dimension-level reasons — the officer sees why, not just what
- Neutral baseline for first-time borrowers — missing history is not treated as bad history

USP of the proposed solution

- Six-pillar score (Revenue, Cash Flow, Transactions, Compliance, Stability, Debt) — a far richer picture than any single metric
- Built on India's DPI stack — consent-based, no proprietary data, aligned with the DPDP Act 2023
- Product recommender maps the score band to CGTMSE / MUDRA / standard loans — more approvals, safely
- Score history gives IDBI post-disbursement monitoring — origination and monitoring from one platform

List of features offered by the solution

Data Aggregation — 5 sources, one consent

- Udyam registry · GST (GSTR-1 / 2A / 2B / 3B, HSN, credit notes, ITC) · ITR (ITR-4 / 5 / 6) · Account Aggregator bank statements (Finvu) · MCA company, charges & DIN-verified directors
- GST ↔ ITR ↔ bank-credit reconciliation with a cross-source anomaly flag for officer review

AI / ML Scoring Engine

- Six weighted pillars, 0–1000: Revenue Vitality 25% · Cash Flow Health 20% · Transaction Trust 15% · Compliance 15% · Business Stability 15% · Debt Serviceability 10%
- Rules engine + ML.NET LightGBM calibration (0.6 / 0.4), SSA cash-flow trend, PFI explanations, neutral baseline when a source is missing

Financial Health Card & Report

- Score gauge, 6-pillar radar, top strengths & risks, lending eligibility — working-capital limit (Nayak) + term-loan capacity, 14-ratio appraisal
- Product recommendation (CGTMSE / MUDRA / standard), Score History trend, printable Financial Report, email delivery

Bank Portal — Customer 360 (new for the Finals)

- Portfolio list with scores & bands, bank dashboard, per-MSME tabs: Score, GST, Bank (AA deep analysis), ITR, MCA, Directors

Hand-off to the lender (new for the Finals)

- Raise a scored MSME as a loan case — IDBI LOS (appraisal case) · ULI (OCEN 4.0 derived data packet) · ONDC (beckn FIS12); the officer previews the exact payload before confirming
- Append-only push audit with score, band and eligibility frozen at the moment the case was raised; pointing a channel at a live endpoint is configuration, not code

Security & Language

- Email-OTP login, CSP security headers, source-IP fraud audit; 7 Indian languages via Bhashini; Groq AI assistant

What's new since Round 1 — built for the Finals

Score History & Trend

- Every analysis is saved (append-only), never overwritten
- Trend chart on the Health Card and in the bank portal
- The base for post-disbursement early warning

Bank Portal · Customer 360

- Portfolio list with scores & bands, bank dashboard
- Per-MSME tabs: Score, GST, Bank, ITR, MCA
- Director drill-down with DIN verification

Loan case → LOS · ULI · ONDC

- Raise a scored MSME as a case from Customer 360
- Preview the exact payload, then confirm — never one click
- A live endpoint is configuration, not a code change

MCA + DIN — 5th data source

- Company status, capital, charges & AGM compliance
- Directors verified against DIN with individual PAN
- Follows entity type — shown only for companies

ITR now inside the score

- Filing timeliness, e-verification, tax paid ratio
- 44AB audit and tax-demand signals
- GST ↔ ITR turnover reconciliation

Lending, Report & Access

- Working-capital limit (Nayak) + term-loan capacity
- 14-ratio appraisal, printable report, email delivery
- 5 seeded demo MSMEs · 7 languages · IDBI palette

Process flow diagram or Use-case diagram

1 · Onboard & log in

1. Enter Udyam number

Udyam registry lookup fetches the enterprise profile

2. Review & consent

Owner confirms details, email + mobile, gives consent and registers (mobile seeded in the Finvu SimBanks sandbox)

3. Email OTP → dashboard

OTP verified, session starts, routed by status

Existing MSME

/Auth/CustLogin → email → OTP → dashboard

2 · Collect data (one button each)

4a. GST

12 months × 7 return files — GSTR-1, 2A, 2B, 3B, HSN, credit notes

4b. Bank statements (AA)

Finvu consent → owner approves → FI fetch of the statement

4c. ITR

Filing, income, tax paid, audit

4d. MCA + DIN

Companies only — status, charges, directors

Stored per MSME

each source saved as JSON in the <UAN>/blob folder; fetched sources are locked

3 · Score

5. Analyze

Five extractors read every file; a missing source redistributes its weight (neutral baseline)

6. Rules 0.6 + LightGBM 0.4

Six dimensions · 0–1000 · band · PFI reasons · GST-ITR-bank anomaly · lending eligibility & products

7. Persist

result.json → Blob · score summary → SQL · score history appended → SQL

Re-analyze

any time — re-runs 5–7 from the stored files; every run adds a history point

4 · Use the score

8. Financial Health Card

Gauge, 6-pillar radar, strengths & risks, products, Score History

9. Financial Report

Printable appraisal — print or email

10. Bank Portal

Bank login → customer list → Customer 360: Score, GST, AA, ITR, MCA, Directors

11. Raise the loan case

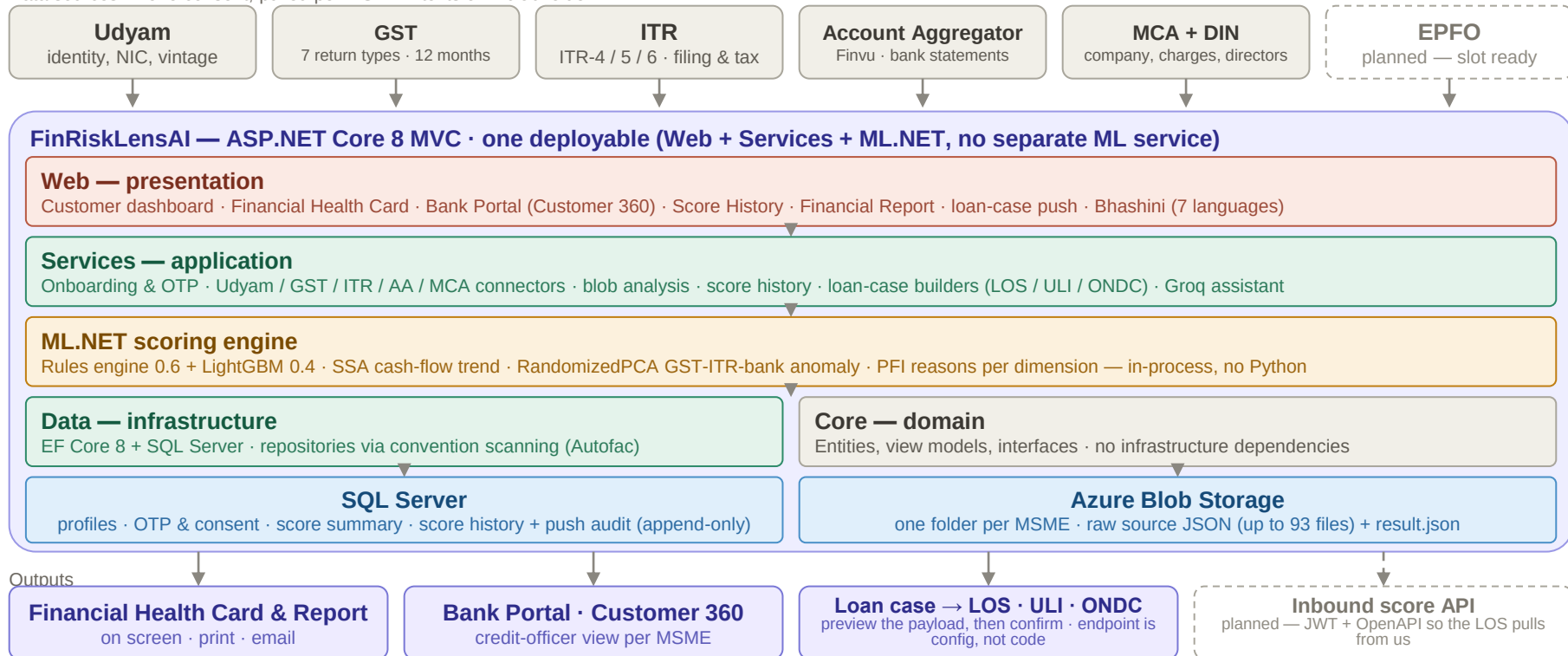
Push to LOS · ULI (OCEN 4.0) · ONDC (beckn) — preview the payload, then confirm

Inbound API — planned

JWT + OpenAPI so the LOS pulls a score from us; the outbound push above is built

Architecture diagram of the proposed solution

Data sources — one consent, pulled per MSME into its own blob folder



Technologies to be used in the solution

Backend — Orchestration & API

- ASP.NET Core 8 MVC (C#) — connectors for Udyam, GST, ITR, Account Aggregator (Finvu), MCA / DIN; outbound loan-case push to LOS / ULI / ONDC
- ML.NET in-process — scoring, feature extraction and explanations in the same deployable; no separate ML service
- Autofac DI with convention scanning · Serilog structured logging · CSP security headers · DataProtection keys persisted

AI / ML Stack

- ML.NET LightGBM calibration — final score = 0.6 rules engine + 0.4 ML model, trained on 3,000 MSME profiles
- ML.NET SSA time-series — 12-month cash-flow trend · RandomizedPCA — GST vs ITR vs bank anomaly detection
- Permutation Feature Importance — per-dimension, audit-ready reasons · Groq LLM — AI assistant for questions on the score

Data & Storage

- SQL Server + EF Core 8 — MSME profiles, OTP & consent, score summary, append-only score history and loan-case push audit; SQL-backed sessions
- Azure Blob Storage — one folder per MSME: raw source JSON (up to 93 files) + result.json

Infrastructure & DevOps

- Single deployable web app (modern monolith) — IIS / Azure App Service / AWS; GitHub source control
- Planned: inbound JWT + OpenAPI spec for LOS score pull; Azure API Management as the ULI / OCEN gateway

Technologies to be used in the solution

Frontend

- ASP.NET Core MVC + Razor views, Bootstrap 5 — server-rendered, responsive; two themes including IDBI's official palette
- Chart.js — score gauge, 6-pillar radar, score-history trend, cash-flow charts · ApexCharts — GST analysis; charts follow the theme switch
- Bhashini website-translation plugin — English + Hindi, Marathi, Tamil, Gujarati, Konkani, Sanskrit

DPI & Data Integrations

- Account Aggregator (Finvu sandbox, SimBanks FIP) · GST System API · Udyam Registry · ITR · MCA 21 (company, charges, DIN)
- Outbound loan case built — IDBI LOS · ULI (OCEN 4.0) · ONDC (beckn FIS12) · inbound DSP score API and EPFO — planned

Security

- Email-OTP customer login and separate bank-admin login, SQL session cache, CSP headers, source-IP fraud / VPN / Tor audit with geolocation

Estimated implementation cost (optional)

₹ 22 L

One-time setup (build + security)

₹ 31 L / yr.

Recurring (cloud + data + support)

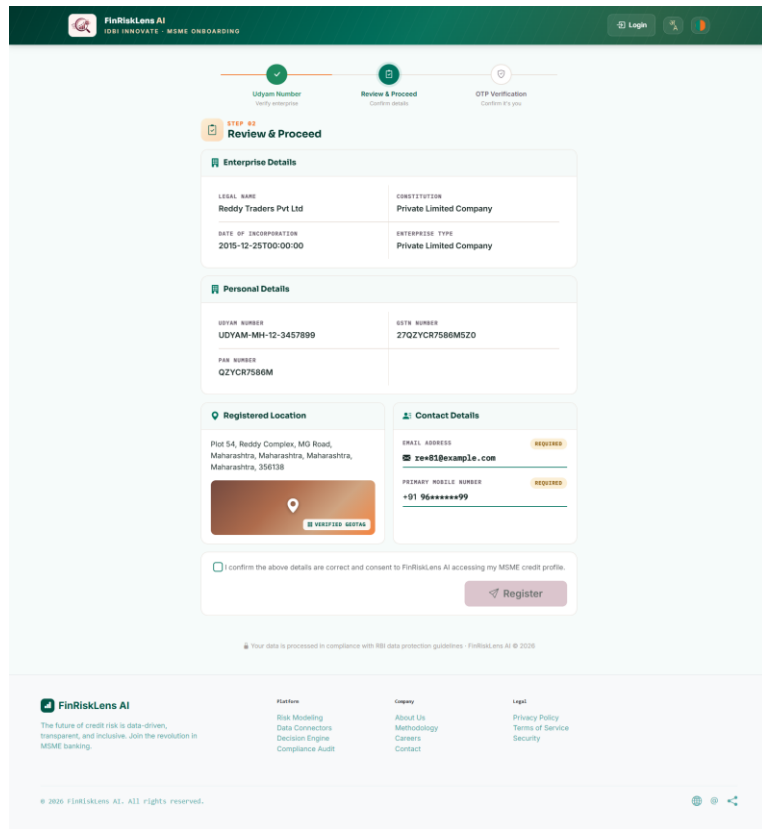
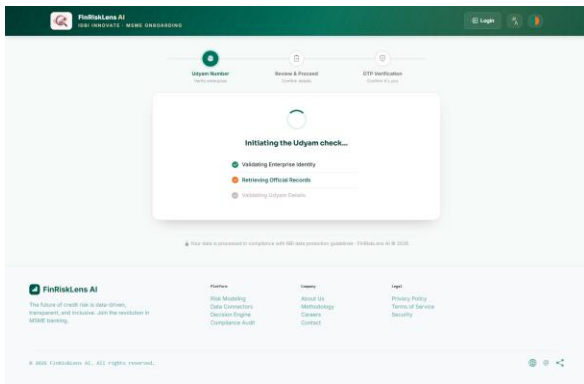
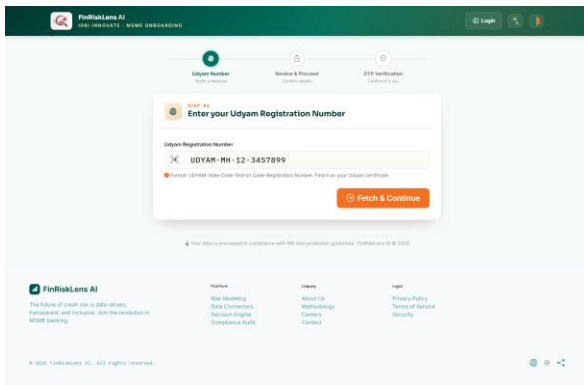
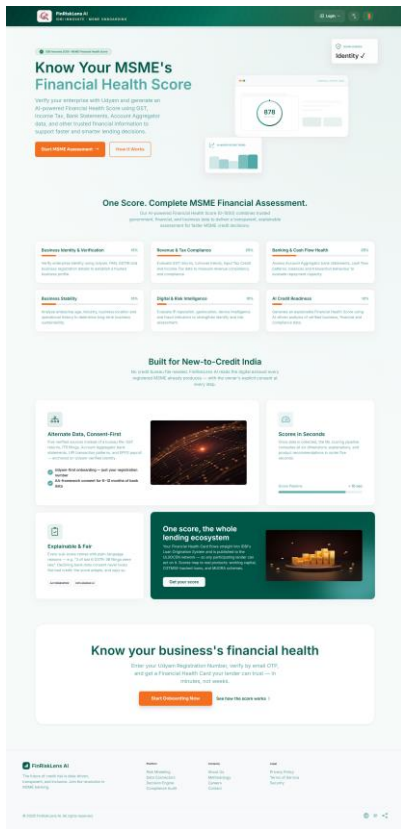
~ ₹ 1.15 Cr

3-year total cost of ownership

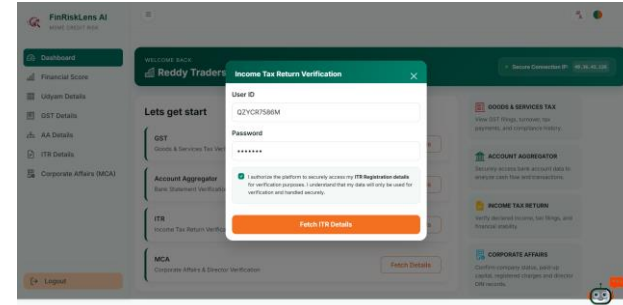
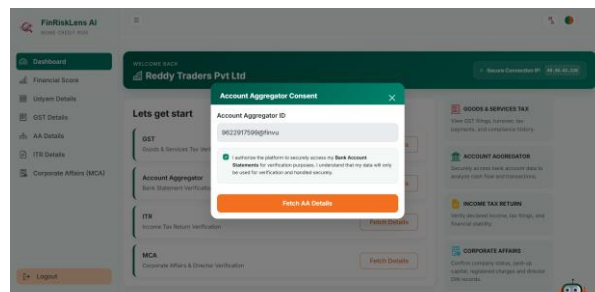
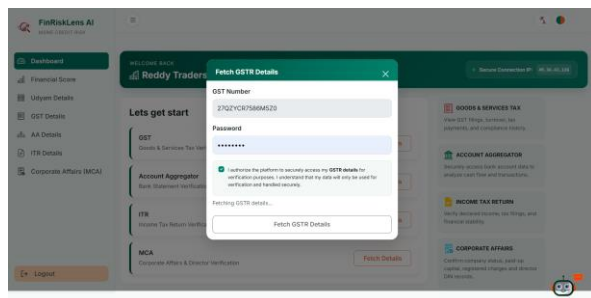
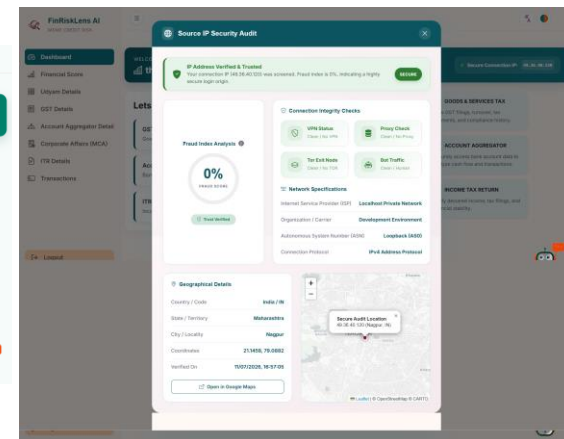
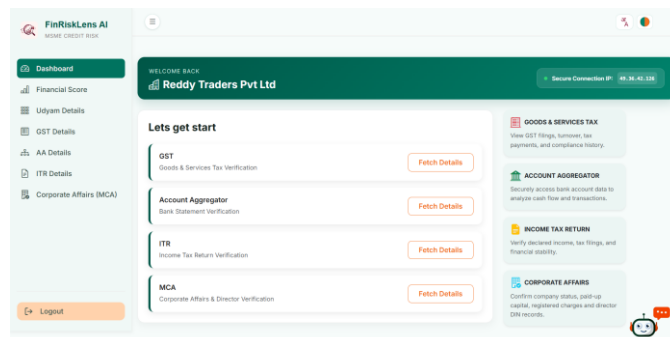
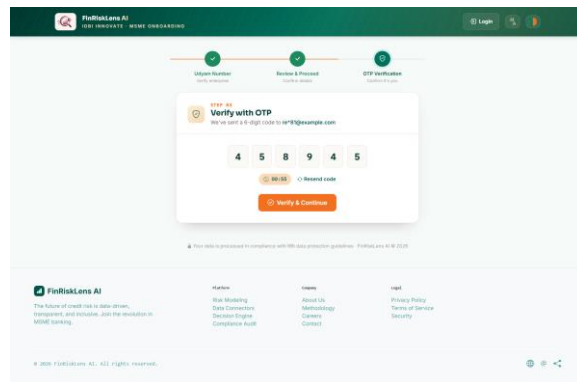
Cost breakdown - centralized cloud deployment serving all IDBI branches pan-India

Cost head	Type	Estimate (INR)
Development - 3-member team, ~6-month build	One-time	₹ 18 L
Security testing & RBI / DPDP compliance audit	One-time	₹ 4 L
Cloud & servers - Azure App Service, SQL, Blob, APIM	Annual	₹ 12 L
Data & API usage - AA (Finvu), GST, ITR	Annual	₹ 10 L
Maintenance, support & ML model retraining	Annual	₹ 9 L

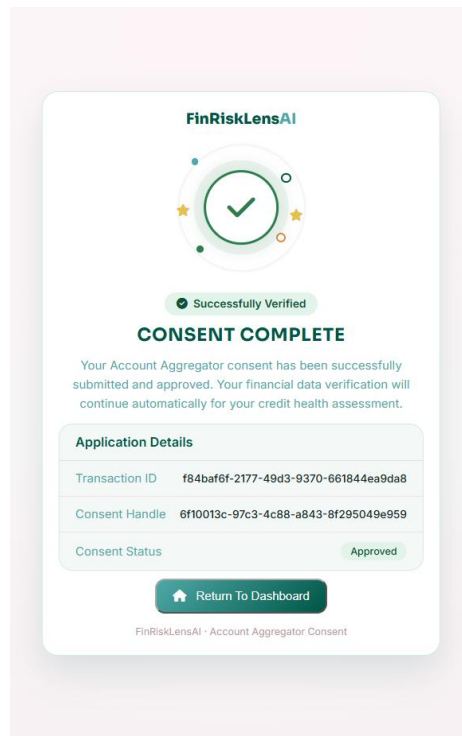
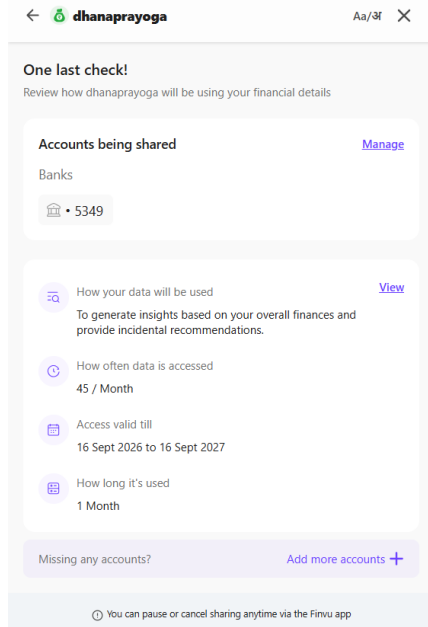
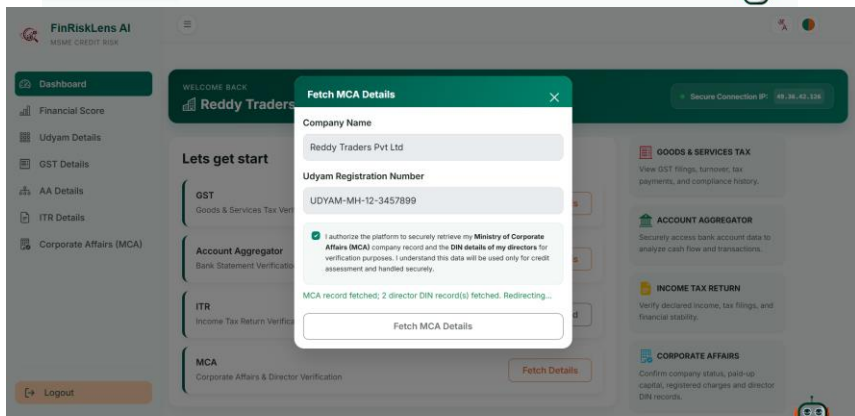
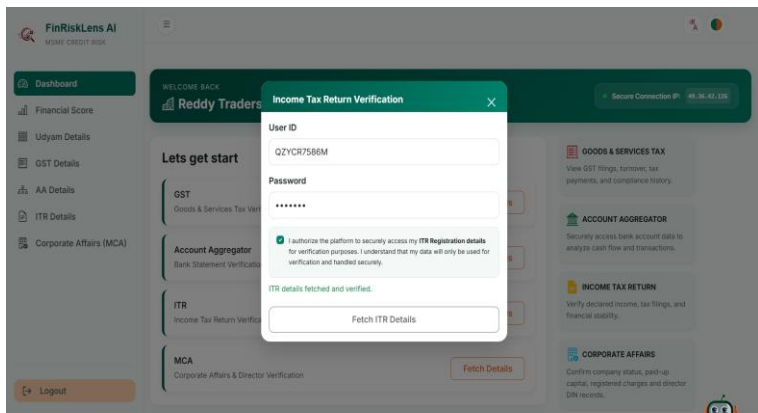
Snapshots of the prototype



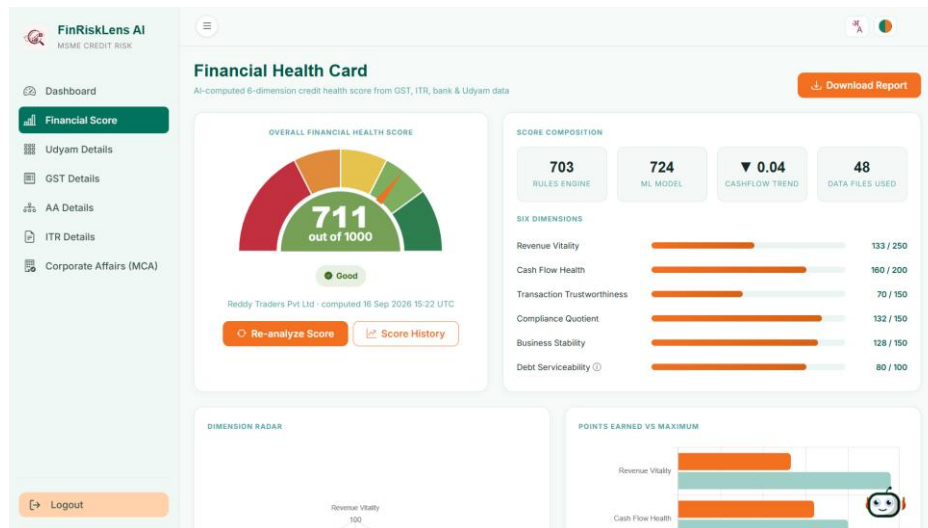
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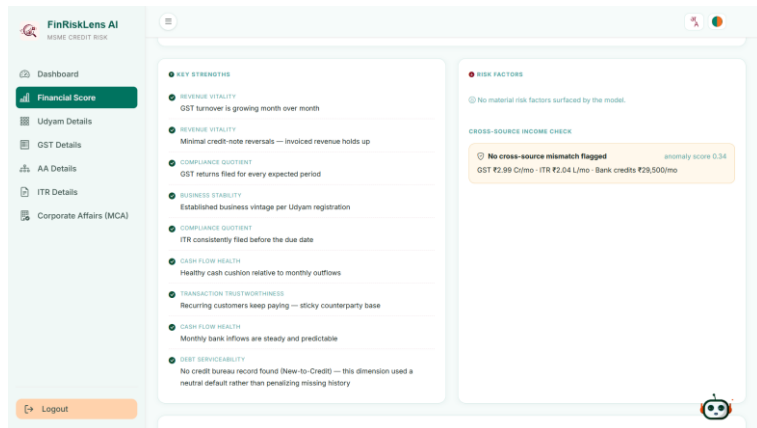
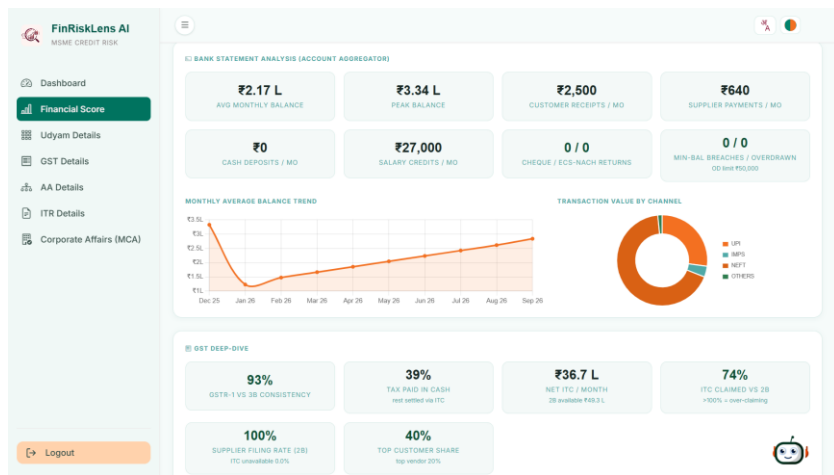
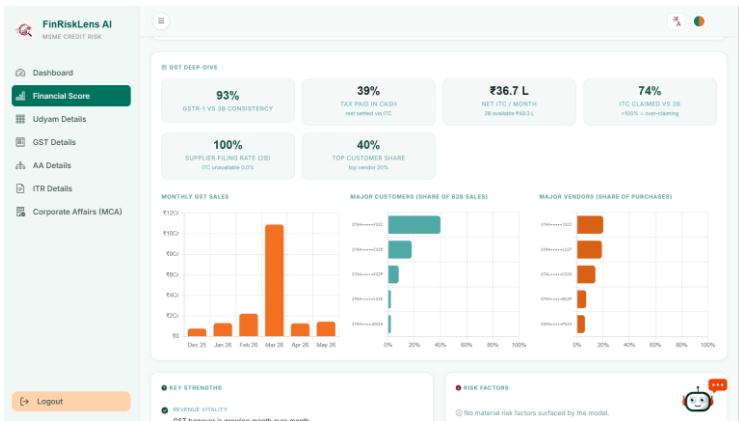
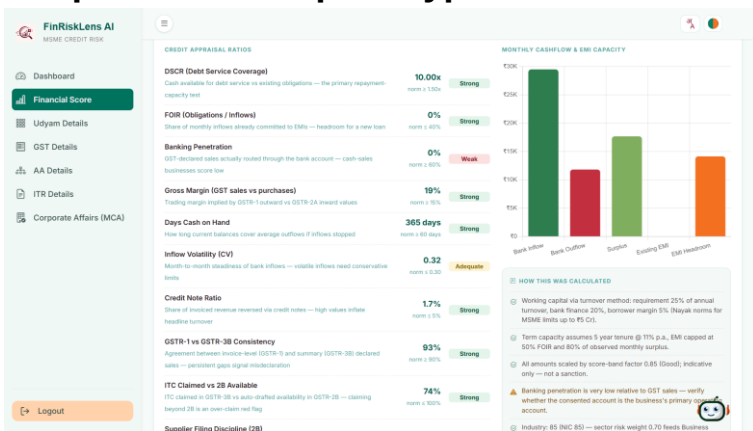
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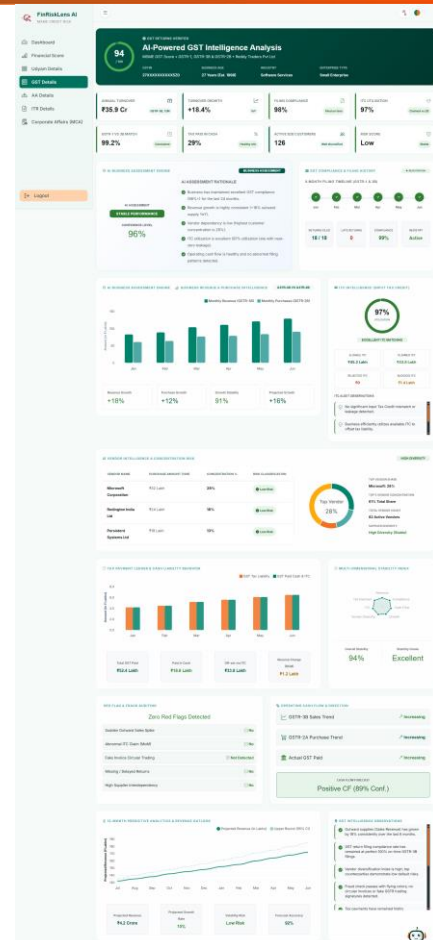
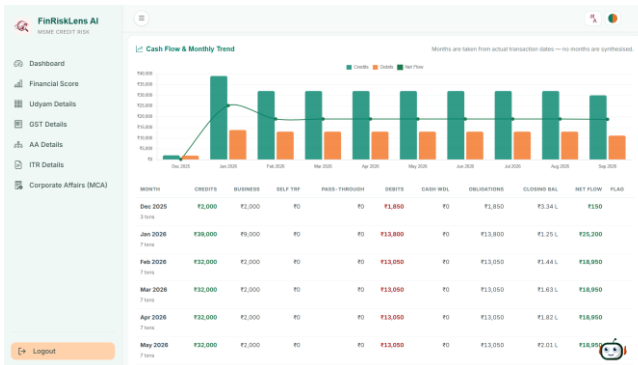
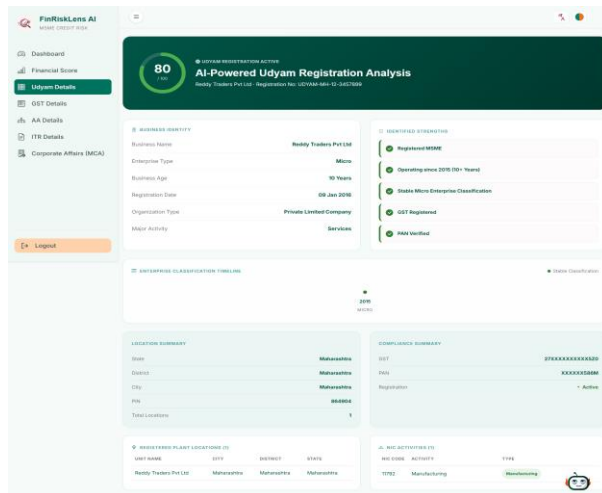
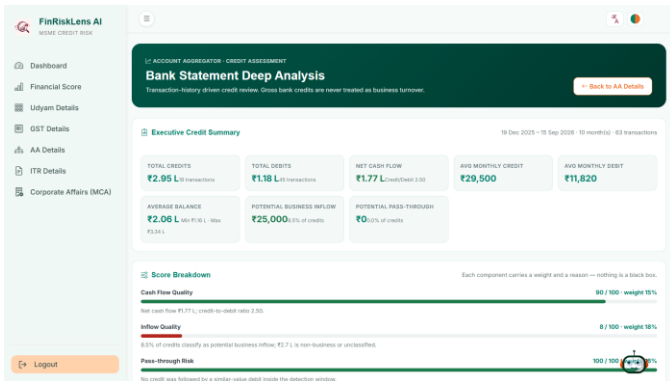
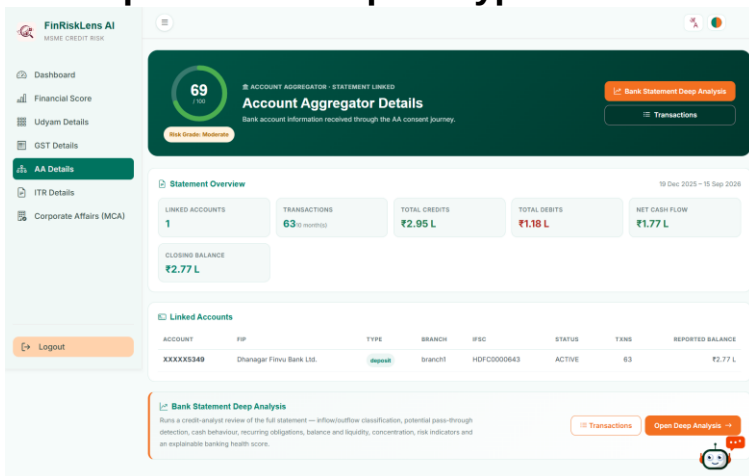
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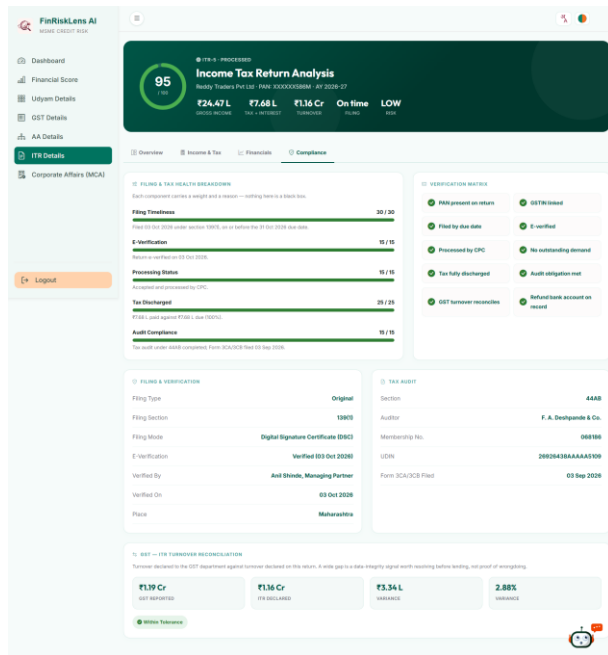


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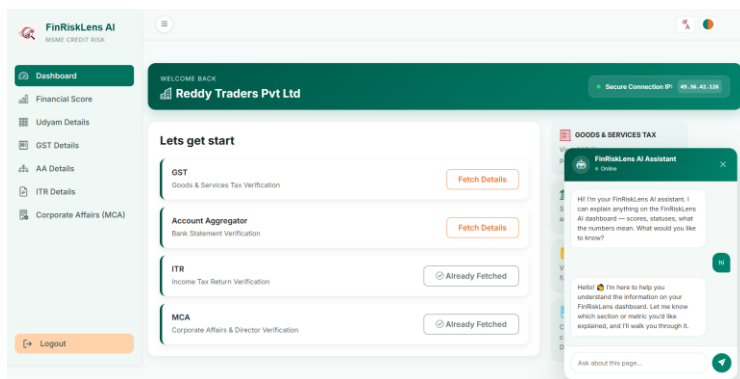
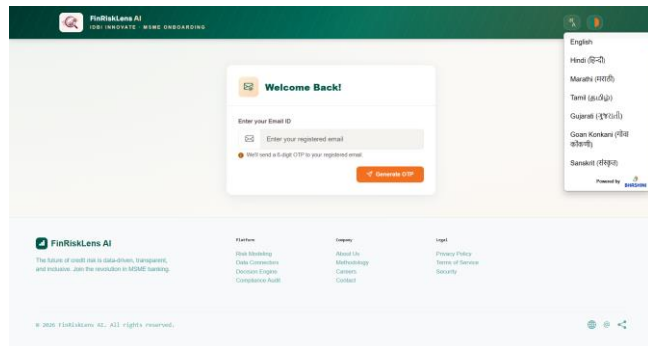
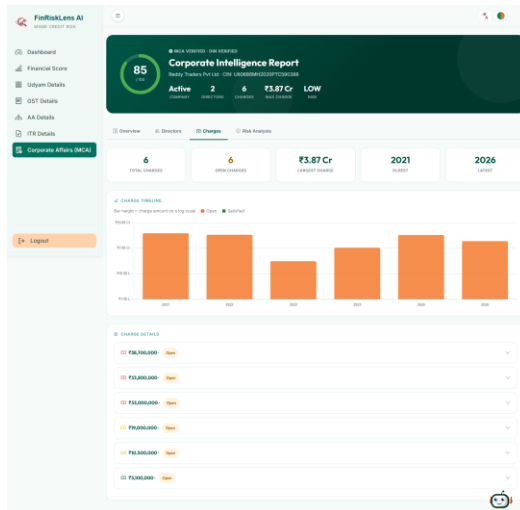
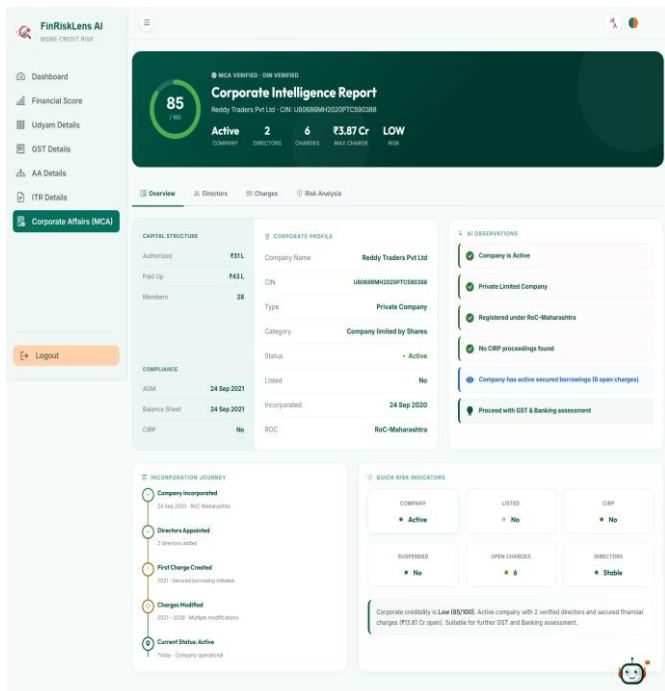


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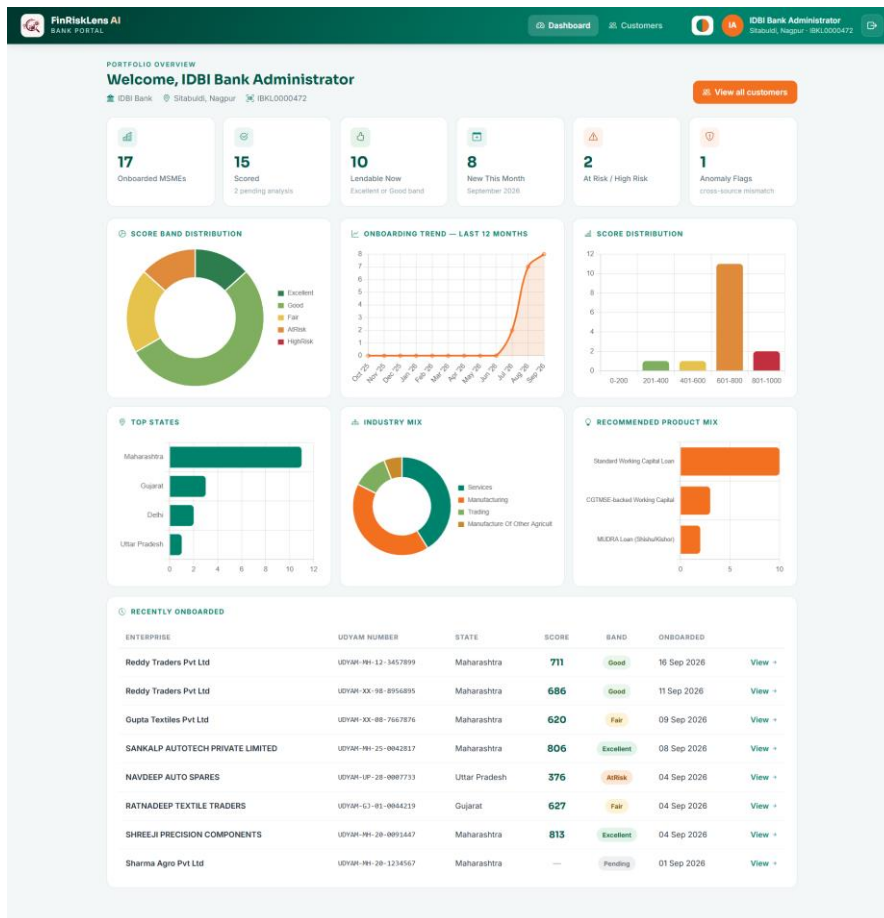
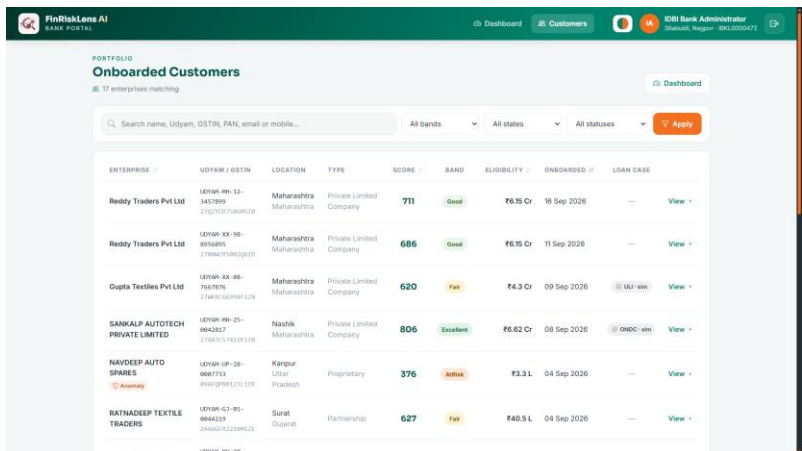
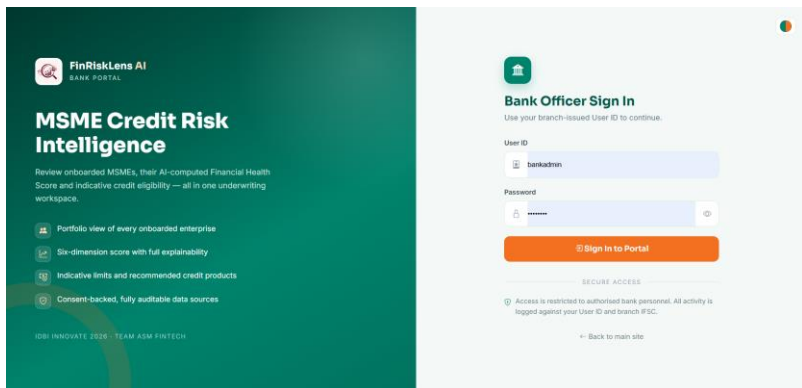




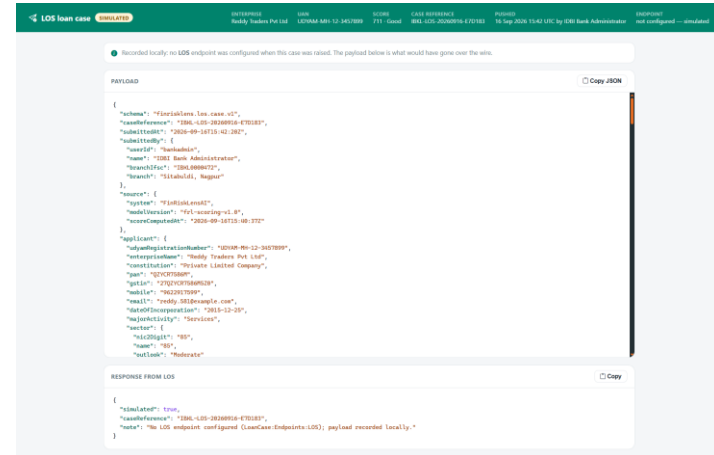
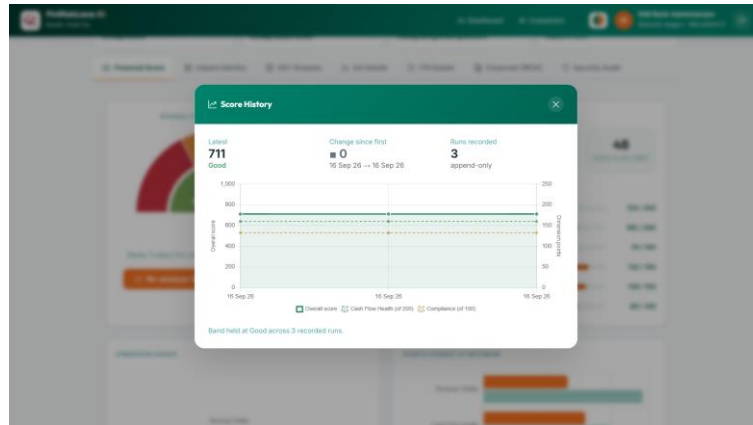
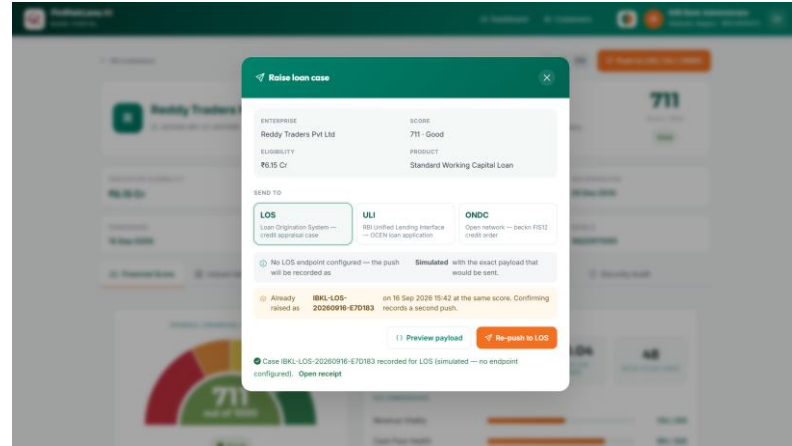
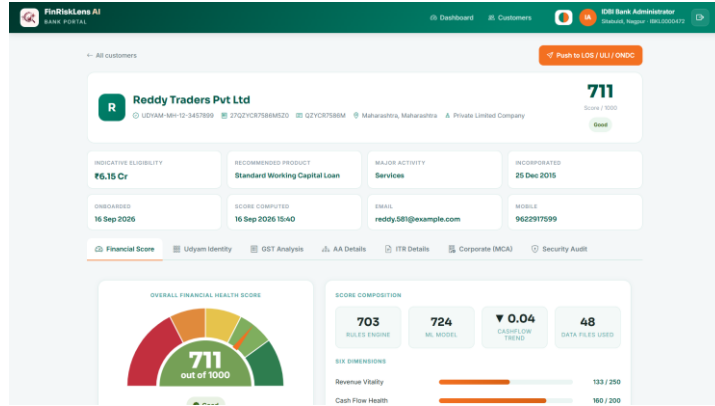
Snapshots of the prototype



Snapshots of the prototype



Snapshots of the prototype



Prototype Performance report/Benchmarking

715/1000

Live score on a real MSME — 12 months of GST + bank data (band: Good)

< 10 s

End-to-end scoring (sub-second when warm)

5

Data sources fused: Udyam · GST · ITR · AA · MCA (+ EPFO ready)

3000

MSME profiles the LightGBM model trains on

Benchmark vs. traditional credit assessment

Metric	Traditional / bureau	FinRiskLensAI
Turnaround time	2-3 weeks (manual)	Seconds (automated)
NTC / NTB borrowers	Rejected - no history	Scored on alternate data
Data sources	1 - bureau file	5 - Udyam, GST, ITR, AA, MCA (+ EPFO ready)
Explainability	Black-box score	PFI reason per dimension
Fraud / integrity check	None	GST-ITR-bank anomaly + MCA / DIN checks
Monitoring	One-time assessment	Score history + trend on every analysis
Hand-off to the lender	Manual file / PDF	LOS, ULI/OCEN or ONDC payload in one action

Additional Details/Future Development (if any)

Next Data Integrations

- EPFO — employee count & PF-remittance regularity (the feature slot is already built)
- GST e-invoice / e-way bill signals and CERSAI collateral lookup
- ONDC / GeM / UPI merchant-settlement data for micro-enterprises

Monitoring, Fraud & Compliance

- Early Warning System on score history — band drop, GST filing lapse, new MCA charge, bounce spike
- Sanctions / PEP screening, RBI fraud registry & wilful-defaulter match
- DPDP consent ledger across all sources with customer-side revoke & auto-purge

Lending Ecosystem & Integration

- Inbound API — JWT + OpenAPI so your LOS pulls a score from us (the outbound case push is already built)
- ULI DSP + OCEN certification and a live ONDC subscriber ID; CIBIL / CRIF bureau blend for NTB borrowers
- GST-invoice financing and co-lending workflows with NBFC partners

Decision & Governance

- Credit-policy console — Approve / Refer / Decline with editable product-wise cut-offs & knock-outs
- PDF Credit Appraisal Memo with data-provenance appendix (source, time, consent ID)
- Model card & governance page (RBI model-risk); portfolio analytics for IDBI teams

Provide links to your:

- GitHub Public Repository : <https://github.com/abhinavmukwane/FinRiskLensAI>
- Demo Video Link (3 Minutes): <https://video.finrisklensai.com> OR [Google Drive Link](#)
- Final Product Link: <https://finrisklensai.com>

Demo accounts — customer login at /Auth/CustLogin (enter email → OTP is shown on screen) · bank portal at /Auth/BankLogin,

shreeji.precision@finrisklens.demo : UDYAM-MH-20-0091447 · 813 Excellent · Shreeji Precision Components

sankalp.autotech@finrisklens.demo : UDYAM-MH-25-0042817 · 806 Excellent · Sankalp Autotech Pvt Ltd (full MCA + DIN)

ratnadeep.textiles@finrisklens.demo : UDYAM-GJ-01-0044219 · 627 Fair · Ratnadeep Textile Traders

navdeep.autospares@finrisklens.demo : UDYAM-UP-28-0007733 · 376 At Risk · Navdeep Auto Spares

This together cover the whole score range — approve, judge, and decline — with every data tab populated.

Bank Login: User ID : **bankadmin** | Password : **Bank@123**



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